# CECH COHOMOLOGY FUNCTORS ON ABELIAN PRESHEAVES FORM A DELTA-FUNCTOR

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*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed. The author's closing sentence omitting the compatibility verification is reproduced as written.

*Editorial note.* The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros; no macro was invented or redefined.

*Editorial note.* The same-chapter pointer is located at its \label line inside the pinned chapter file; the two cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

*Allowed standard prerequisites (editorial).*

- Cohomological delta-functors. Homology, Definition labelled homology-definition-cohomological-delta-functor (canonical Stacks tag 010Q); named in the counted statement itself. Supplies the structure that the Cech cohomology functors must be shown to carry, i.e. the boundary maps and their compatibility.
- The boundary maps of the long exact sequence of a short exact sequence of cochain complexes. Homology, Lemma labelled homology-lemma-long-exact-sequence-cochain (canonical Stacks tag 0117); invoked at proof line 960. Supplies the existence of the connecting maps and of the long exact sequence; the functoriality of the construction is the counted step.

*Source reference.* Original lemma 03AQ: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 926–926. Same-chapter pointer invoked at proof line 955 for exactness of the Cech complex functor; the construction of the connecting maps on top of it is retained verbatim in the counted span.

*Source reference.* Original definition 010Q: sources/stacks-chow/tags.txt, source lines 1336–1336. Cross-chapter pointer named in the counted statement at line 950 for the notion of a cohomological delta-functor; the construction of the boundary maps is retained in the counted span.

*Source reference.* Original lemma 0117: sources/stacks-chow/tags.txt, source lines 1353–1353. Cross-chapter standard prerequisite invoked once at proof line 960 to obtain the connecting homomorphisms; the surrounding argument that these give a delta-functor is retained in the counted span.

**Lemma 0.1.** *Let $\mathcal{C}$ be a category. Let $\mathcal{U} = \{U_i \to U\}_{i \in I}$ be a family of morphisms with fixed target such that all fibre products $U_{i_0} \times_U \ldots \times_U U_{i_p}$ exist in $\mathcal{C}$. The*

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*functors $\mathcal{F} \mapsto \check{H}^n(\mathcal{U}, \mathcal{F})$ form a $\delta$-functor from the abelian category $PAb(\mathcal{C})$ to the category of $\mathbf{Z}$-modules (see Homology, Definition 010Q).*

*Proof.* By Lemma 03AQ a short exact sequence of abelian presheaves $0 \to \mathcal{F}_1 \to \mathcal{F}_2 \to \mathcal{F}_3 \to 0$ is turned into a short exact sequence of complexes of $\mathbf{Z}$-modules. Hence we can use Homology, Lemma 0117 to get the boundary maps $\delta_{\mathcal{F}_1 \to \mathcal{F}_2 \to \mathcal{F}_3} : \check{H}^n(\mathcal{U}, \mathcal{F}_3) \to \check{H}^{n+1}(\mathcal{U}, \mathcal{F}_1)$ and a corresponding long exact sequence. We omit the verification that these maps are compatible with maps between short exact sequences of presheaves. □